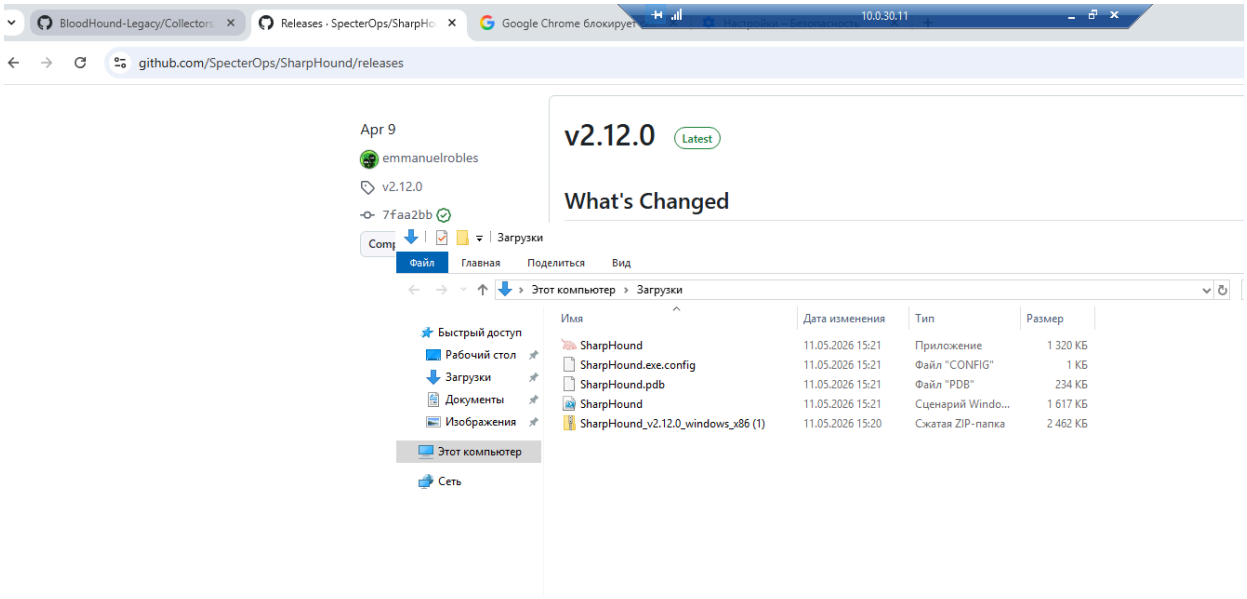
1. Сбор данных с помощью SharpHound

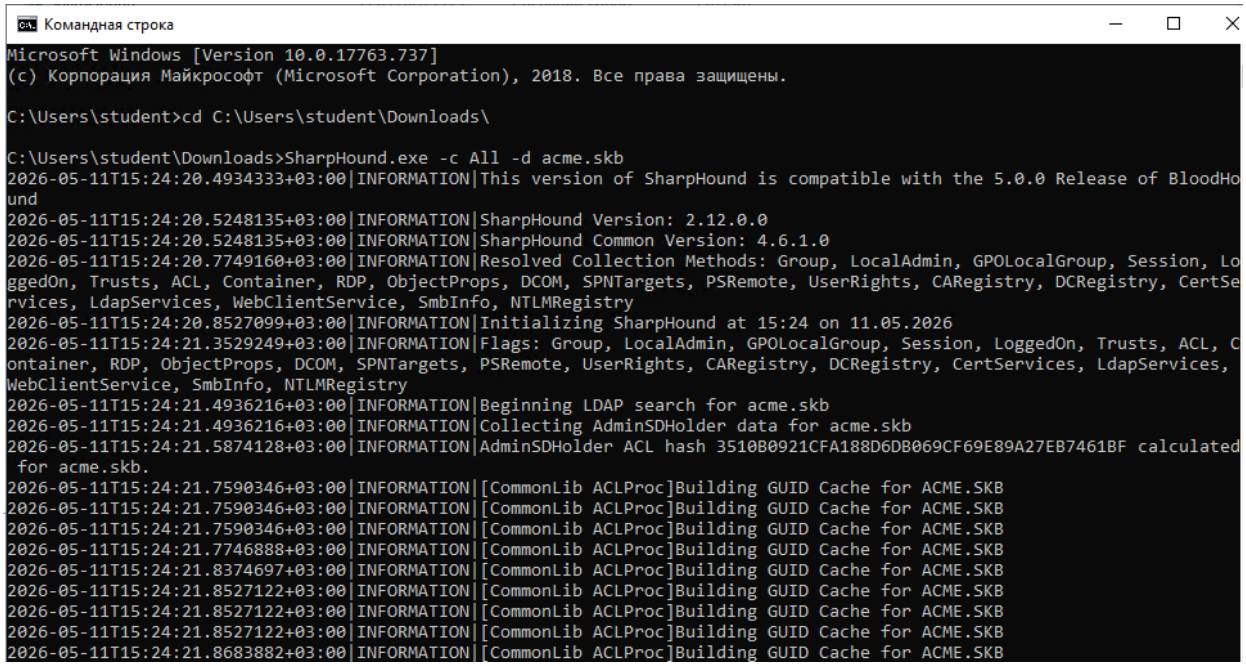
Скачивание SharpHound на ws01

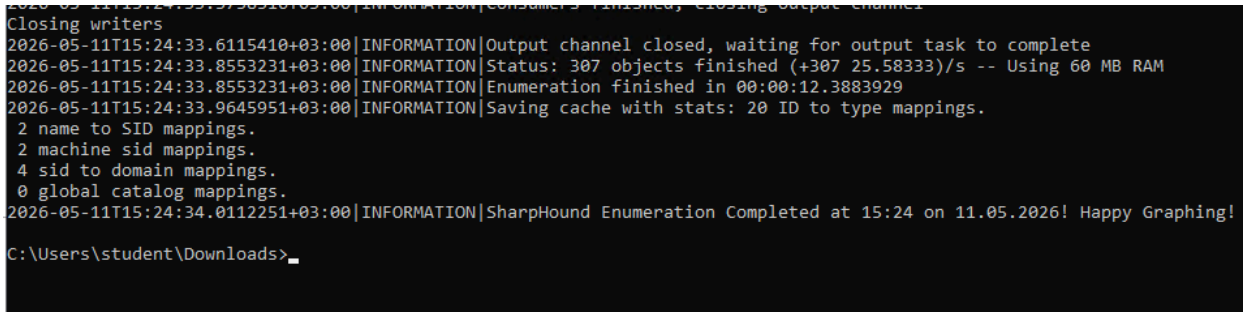
На машине ws01 с GitHub скачиваем последний релиз SharpHound



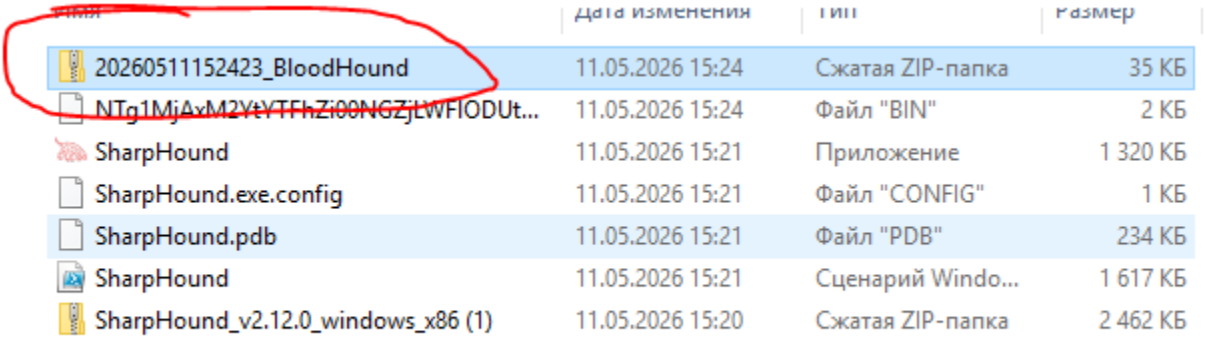
Запуск SharpHound

С явным указанием домена:

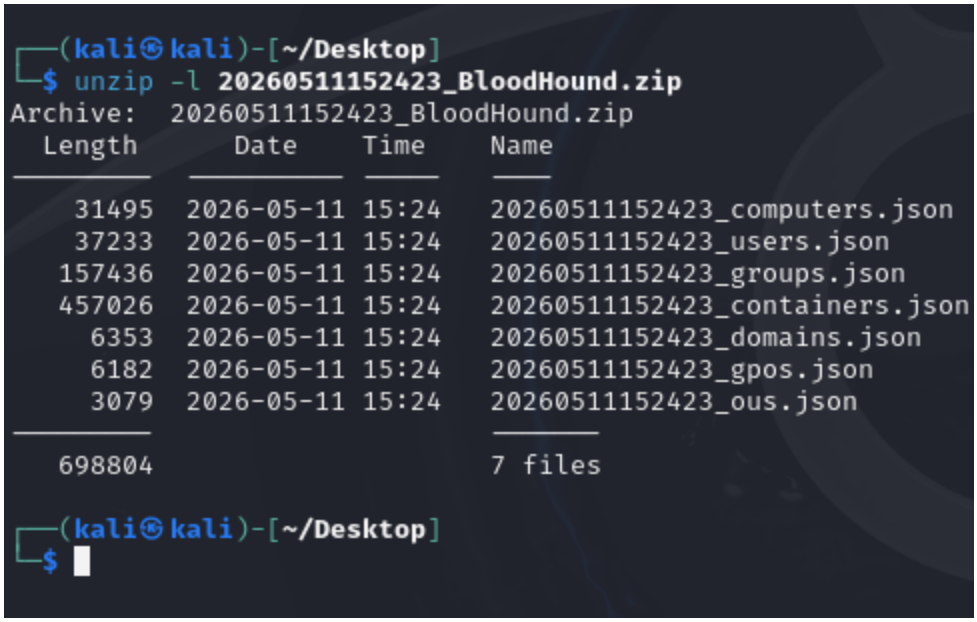




ZIP-архив с результатами готов.



Копируем архив себе на машину Kali и удостоверяемся, что в нем есть нужные файлы



Этап 2. Установка и настройка BloodHound

Устанавливаем BloodHound

```
┌──(kali㉿kali)-[~]
└─$ sudo apt update
[sudo] password for kali:
Get:1 http://mirror.krfoss.org/kali kali-rolling InRelease [34.0 kB]
Get:2 http://mirror.krfoss.org/kali kali-rolling/main amd64 Packages [20.9 MB]
Get:3 http://mirror.krfoss.org/kali kali-rolling/main amd64 Contents (deb) [53.0 MB]
Get:4 http://mirror.krfoss.org/kali kali-rolling/contrib amd64 Packages [117 kB]
Get:5 http://mirror.krfoss.org/kali kali-rolling/contrib amd64 Contents (deb) [261 kB]
Get:6 http://mirror.krfoss.org/kali kali-rolling/non-free amd64 Packages [188 kB]
Get:7 http://mirror.krfoss.org/kali kali-rolling/non-free amd64 Contents (deb) [906 kB]
Fetched 75.4 MB in 20s (3,716 kB/s)
1376 packages can be upgraded. Run 'apt list --upgradable' to see them.

┌──(kali㉿kali)-[~]
└─$ sudo apt install bloodhound -y
The following packages were automatically installed and are no longer required:
  cpp-12                  libblockdev-part2       libimobiledevice6           libpostproc56
  debtags                 libblockdev-swap2       libjavascriptcoregtk-4.0-18 libpython3.10-mi
  gdisk                   libblockdev-utils2      libjim0.81                  libpython3.10-st
  gsfonts                 libblockdev2            libkate1                    libpython3.12-mi
  kali-debtags            libcairomm-1.0-1v5      libllvm14                   libpython3.12-st
  libabsl20220623         libcanberra-gtk-module  libllvm15                   libqt5ct-common1
  libadwaita-1-0          libcanberra-gtk0        libmagickcore-6.q16-6       libssh-gcrypt-4
  libappstream5           libcbor0.8              libmagickcore-6.q16-6-extra libstemmer0d
  libapt-pkg6.0t64        libcrypt-dev            libmagickwand-6.q16-6       libswscale6
  libassuan0              libgail-common          libmpdec3                   libtiff5
  libatk-adaptor          libgail18t64            libnfs13                    libusbmuxd6
  libatkmm-1.6-1v5        libgspell-1-2           libpangomm-1.4-1v5          libutempter0
  libavfilter8            libgtk2.0-0t64          libpcre3                    libutf8proc2
  libavformat59           libgtk2.0-bin           libperl5.36                 libwebkit2gtk-4.
  libavif15               libgtk2.0-common        libplacebo208               libwpe-1.0-1
  libblockdev-crypto2     libgtkmm-3.0-1v5        libplist3                   libwpebackend-fd
  libblockdev-loop2       libgupnp-igd-1.0-4      libpoppler123               libxmlb2
  libblockdev-part-err2   libicu71                libpoppler126               libzxing2
Use 'sudo apt autoremove' to remove them.

Installing:
  bloodhound

Installing dependencies:
  azurehound  binfmt-support  fastjar  jarwrapper  neo4j  openjdk-11-jre  openjdk-11-jre-headl

Suggested packages:
  fonts-ipafont-gothic  fonts-ipafont-mincho  fonts-wqy-microhei  | fonts-wqy-zenhei  fonts-ir

Summary:
  Upgrading: 0, Installing: 9, Removing: 0, Not Upgrading: 1376
  2 not fully installed or removed.
  Download size: 260 MB
  Space needed: 651 MB / 4,600 MB available
```

Устанавливаем и запускаем СУБД neo4j

```
┌──(kali㉿kali)-[~]
└─$ sudo neo4j start
[sudo] password for kali:
Directories in use:
home:         /usr/share/neo4j
config:       /usr/share/neo4j/conf
logs:         /etc/neo4j/logs
plugins:      /usr/share/neo4j/plugins
import:       /usr/share/neo4j/import
data:         /etc/neo4j/data
certificates: /usr/share/neo4j/certificates
licenses:     /usr/share/neo4j/licenses
run:          /var/lib/neo4j/run
Starting Neo4j.
Started neo4j (pid:3946). It is available at http://localhost:7474
There may be a short delay until the server is ready.

┌──(kali㉿kali)-[~]
└─$
```

Удостоверяемся, что neo4j работает как надо:

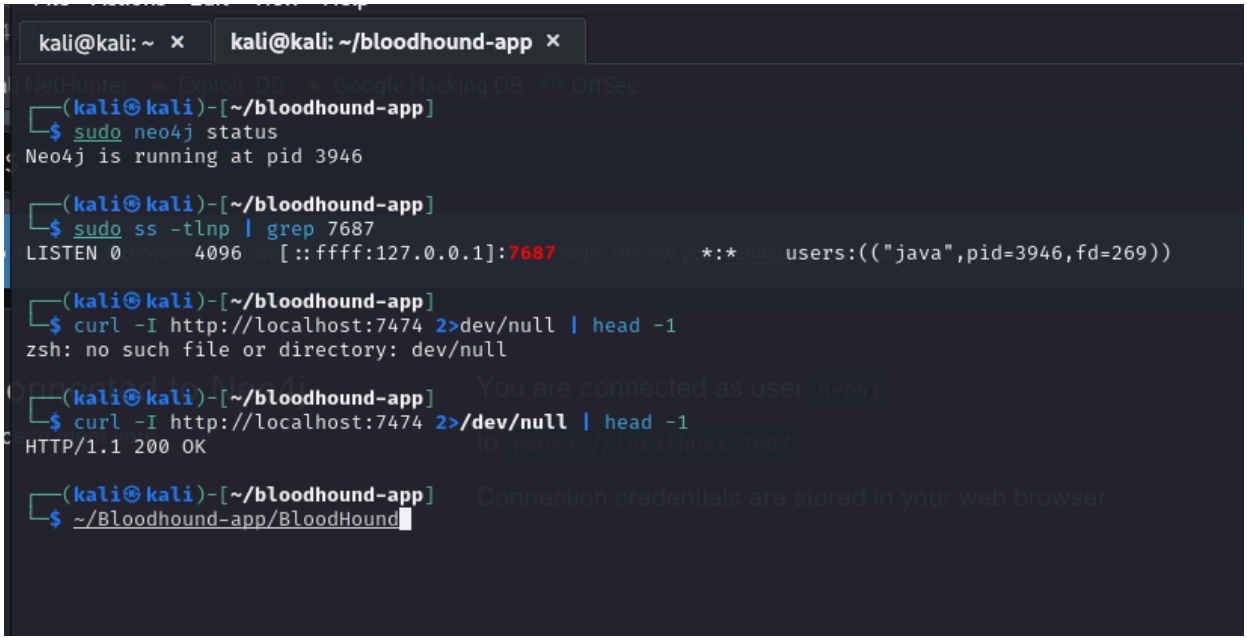
```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl enable neo4j --now
Synchronizing state of neo4j.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable neo4j
Created symlink '/etc/systemd/system/multi-user.target.wants/neo4j.service' → '/usr/lib/systemd/system/neo4j.service'.

┌──(kali㉿kali)-[~]
└─$ sudo neo4j-admin dbms set-initial-password bloodhound123
Changed password for user 'neo4j'. IMPORTANT: this change will only take effect if performed before the database is starte
d for the first time.

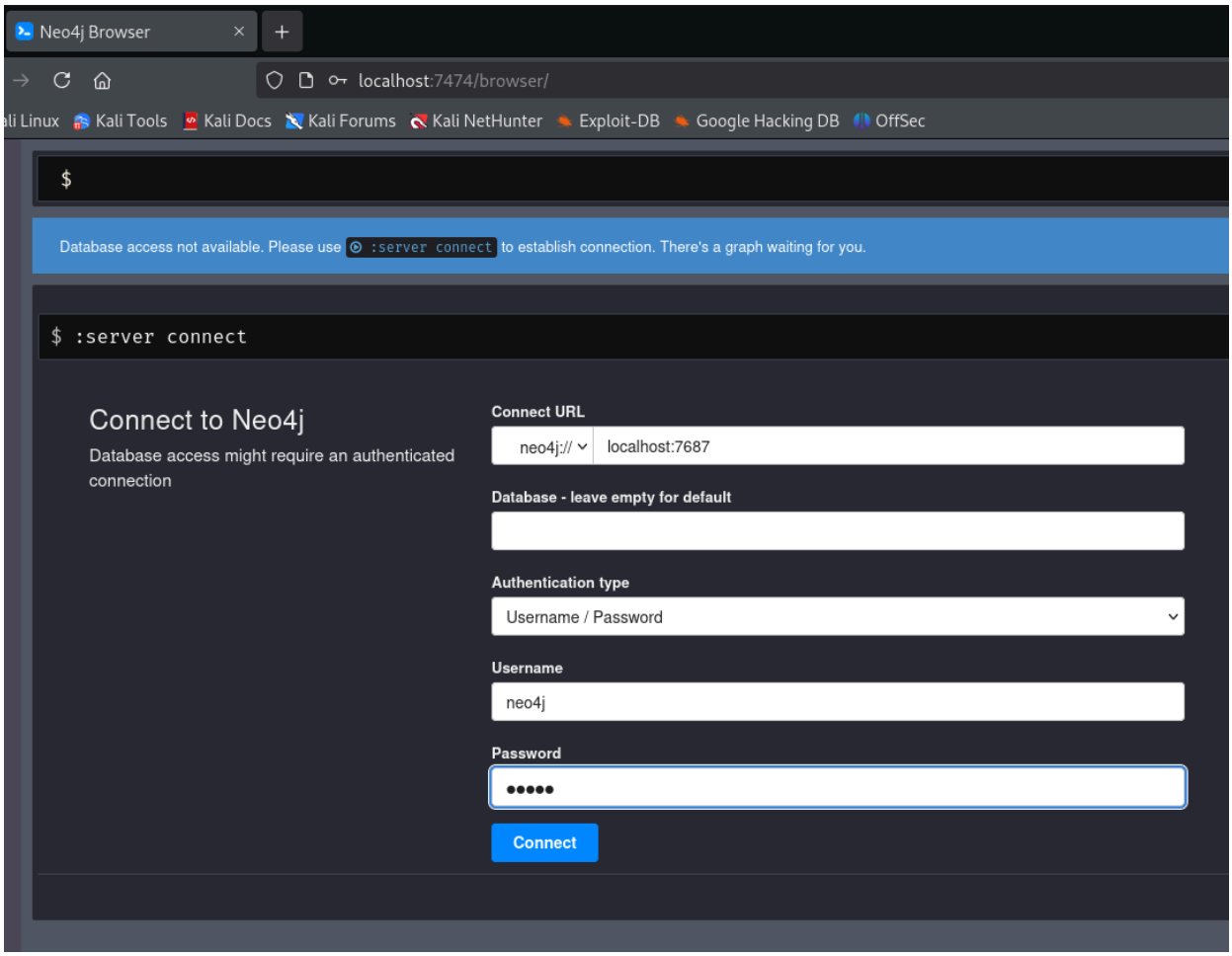
┌──(kali㉿kali)-[~]
└─$ sudo systemctl restart neo4j

┌──(kali㉿kali)-[~]
└─$ sudo systemctl status neo4j
● neo4j.service - Neo4j Graph Database
     Loaded: loaded (/usr/lib/systemd/system/neo4j.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-05-11 10:15:14 EDT; 33s ago
 Invocation: 39e1054a08014c3bb23779f2ea4ec2c8
   Main PID: 7996 (java)
      Tasks: 95 (limit: 18948)
     Memory: 770.5M (peak: 783.4M)
        CPU: 18.552s
     CGroup: /system.slice/neo4j.service
             ├─7996 /usr/bin/java -Xmx128m -XX:+IgnoreUnrecognizedVMOptions --sun-misc-unsafe-memory-access=allow -classp>
             └─8042 /usr/lib/jvm/java-21-openjdk-amd64/bin/java -cp "/var/lib/neo4j/plugins/*:/etc/neo4j/*:/usr/share/neo>

May 11 10:15:19 kali neo4j[8042]: 2026-05-11 14:15:19.048+0000 WARN  SECURITY WARNING: X-Forward headers accept any hostn>
May 11 10:15:20 kali neo4j[8042]: 2026-05-11 14:15:20.373+0000 INFO  Anonymous Usage Data is being sent to Neo4j, see htt>
May 11 10:15:20 kali neo4j[8042]: 2026-05-11 14:15:20.911+0000 INFO  Bolt enabled on localhost:7687.
May 11 10:15:21 kali neo4j[8042]: 2026-05-11 14:15:21.376+0000 INFO  This server has built-in support for online monitori>
May 11 10:15:22 kali neo4j[8042]: 2026-05-11 14:15:22.114+0000 INFO  HTTP enabled on localhost:7474.
May 11 10:15:22 kali neo4j[8042]: 2026-05-11 14:15:22.115+0000 INFO  Remote interface available at http://localhost:7474/
May 11 10:15:22 kali neo4j[8042]: 2026-05-11 14:15:22.119+0000 INFO  id: 5A8CE3EA581612C834AAD50681933E1ACE47997A9EE7FAC2>
May 11 10:15:22 kali neo4j[8042]: 2026-05-11 14:15:22.120+0000 INFO  name: system
May 11 10:15:22 kali neo4j[8042]: 2026-05-11 14:15:22.120+0000 INFO  creationDate: 2026-05-11T14:14:12.403Z
May 11 10:15:22 kali neo4j[8042]: 2026-05-11 14:15:22.120+0000 INFO  Started.
lines 1-22/22 (END)
```



В браузере устанавливаем пароль для Neo4j



Запуск BloodHound



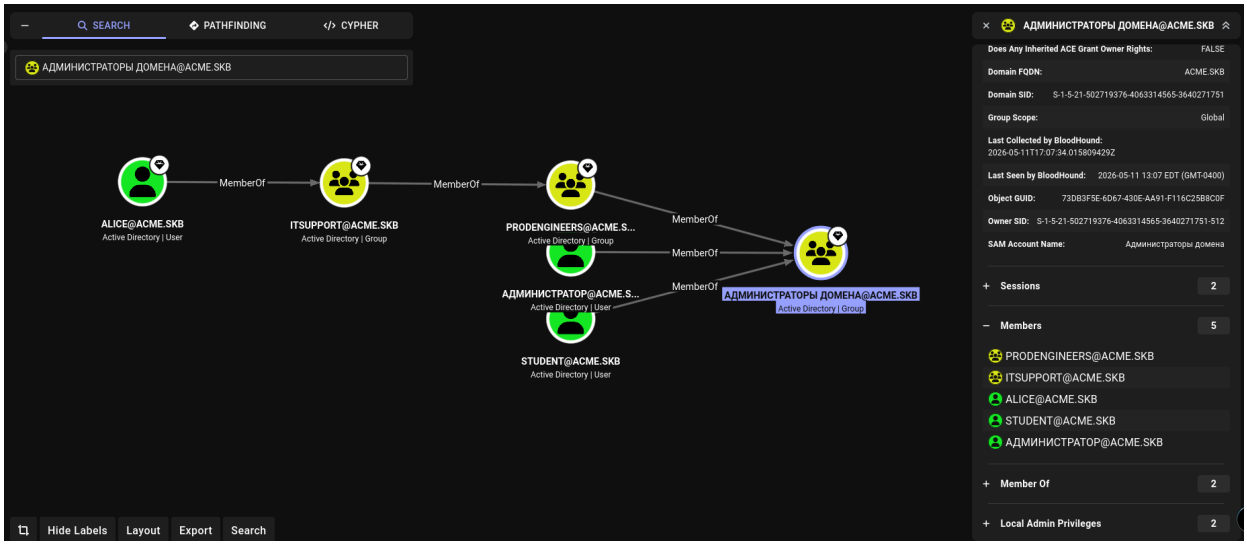


В обычном режиме выдает ошибку и в инфраструктуре, и на локальной ВМ, поэтому запускаем с флагами: ./BloodHound --no-sandbox --disable-gpu

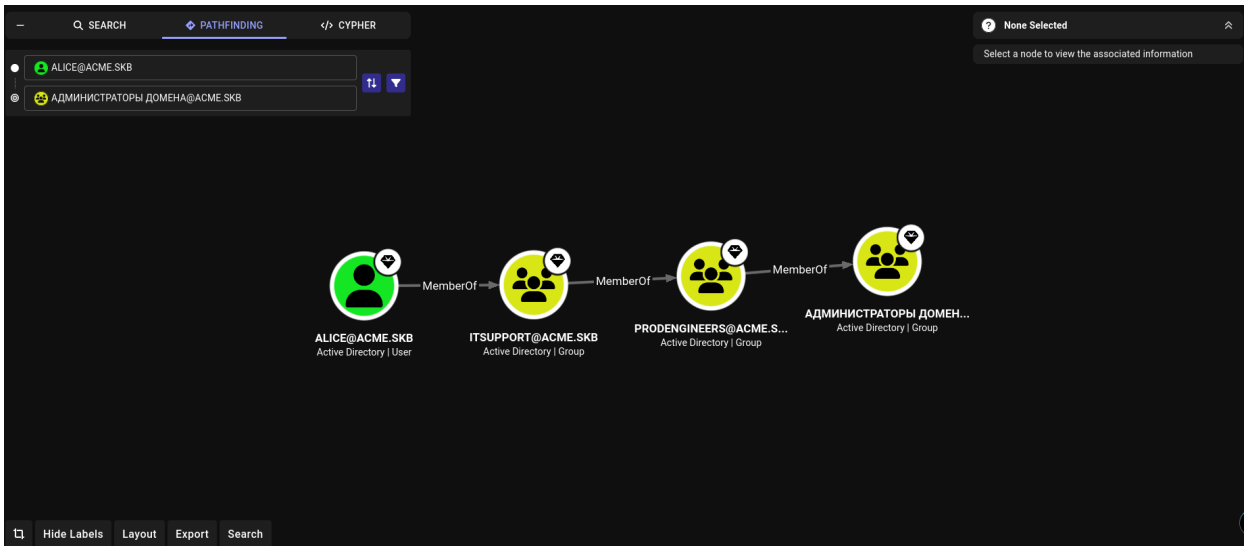
```
┌──(kali㉿kali)-[~/bloodhound-app/BloodHound-linux-x64]
└─$ ./BloodHound
(node:20108) electron: The default of contextIsolation is deprecated and will be changing from false to tr
 See https://github.com/electron/electron/issues/23506 for more information
[20108:0511/155826.054794:FATAL:gpu_data_manager_impl_private.cc(445)] GPU process isn't usable. Goodbye.
zsh: trace trap  ./BloodHound

┌──(kali㉿kali)-[~/bloodhound-app/BloodHound-linux-x64]
└─$

┌──(kali㉿kali)-[~/bloodhound-app/BloodHound-linux-x64]
└─$ ./BloodHound --no-sandbox --disable-gpu --disable-software-rasterizer
```



Состав группы Domain Admins (АДМИНИСТРАТОРЫ ДОМЕНА@ACME.SKB) и связи членства



Путь атаки от пользователя Alice к группе Domain Admins через вложенные группы
Alice → ITSUPPORT → PRODENGINEERS → Domain Admins
Уязвимость: непрямое членство в привилегированной группе через цепочку вложенности.